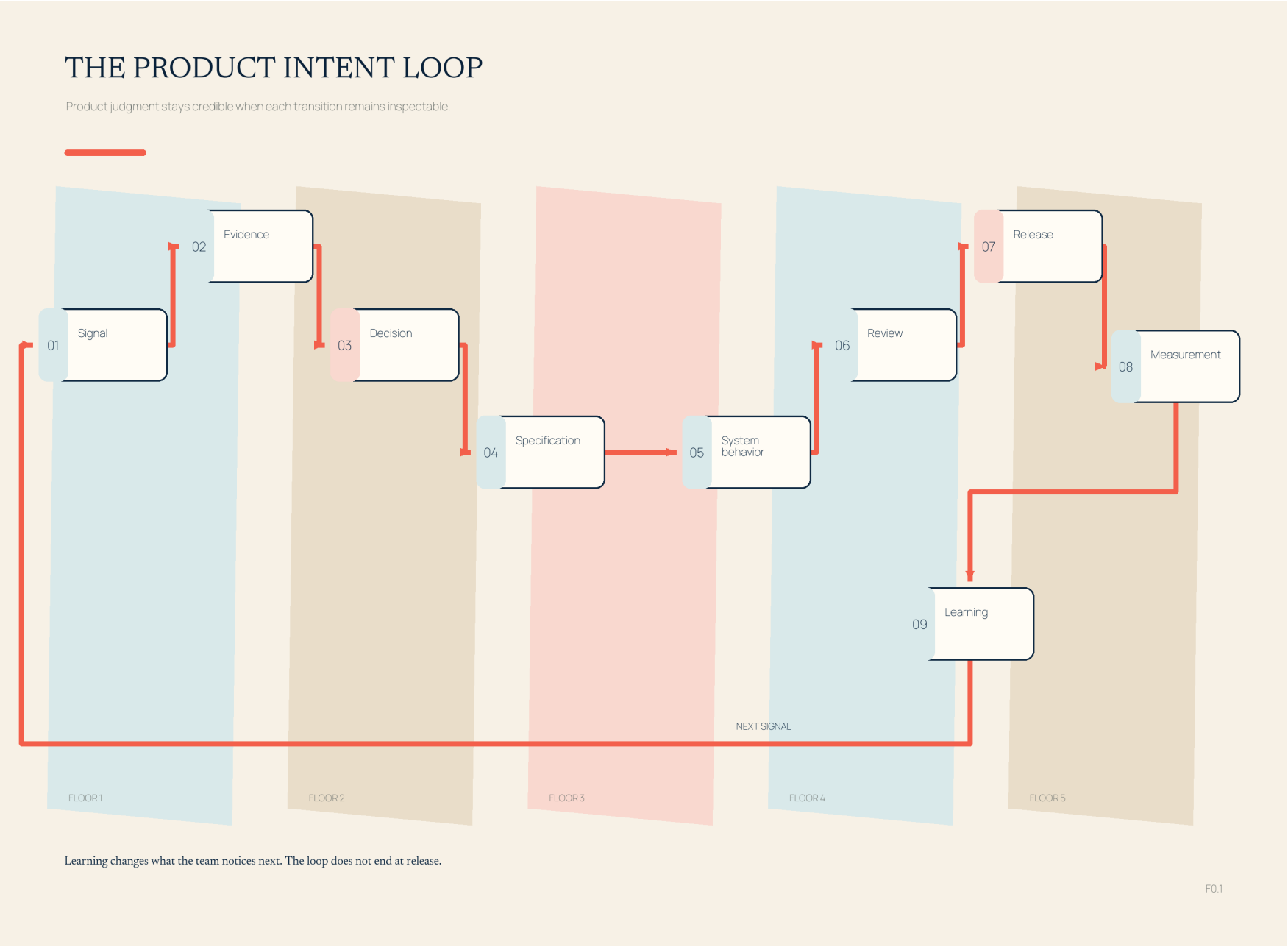


Figure 0.1. Signal-to-learning operating chain



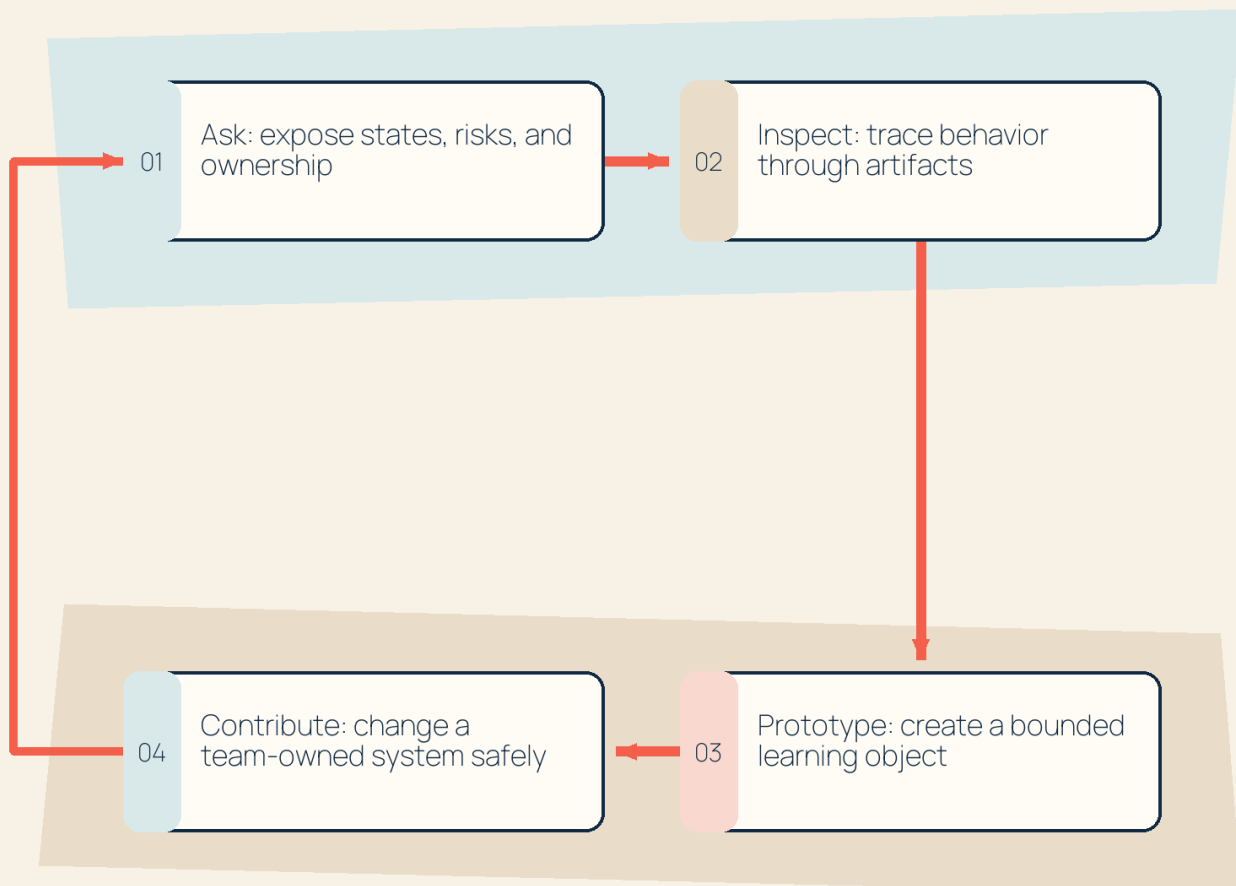
Product intent survives only when each transition remains inspectable; learning begins the next loop rather than ending the work.

Figure 1.1. Participation ladder: ask, inspect, prototype, contribute

THE TECHNICAL PARTICIPATION CLIMB

Technical confidence grows through observable contribution, not a change in title.

EXPERIENCE SHARPENS THE NEXT QUESTION



Move only as far as the product question and risk require.

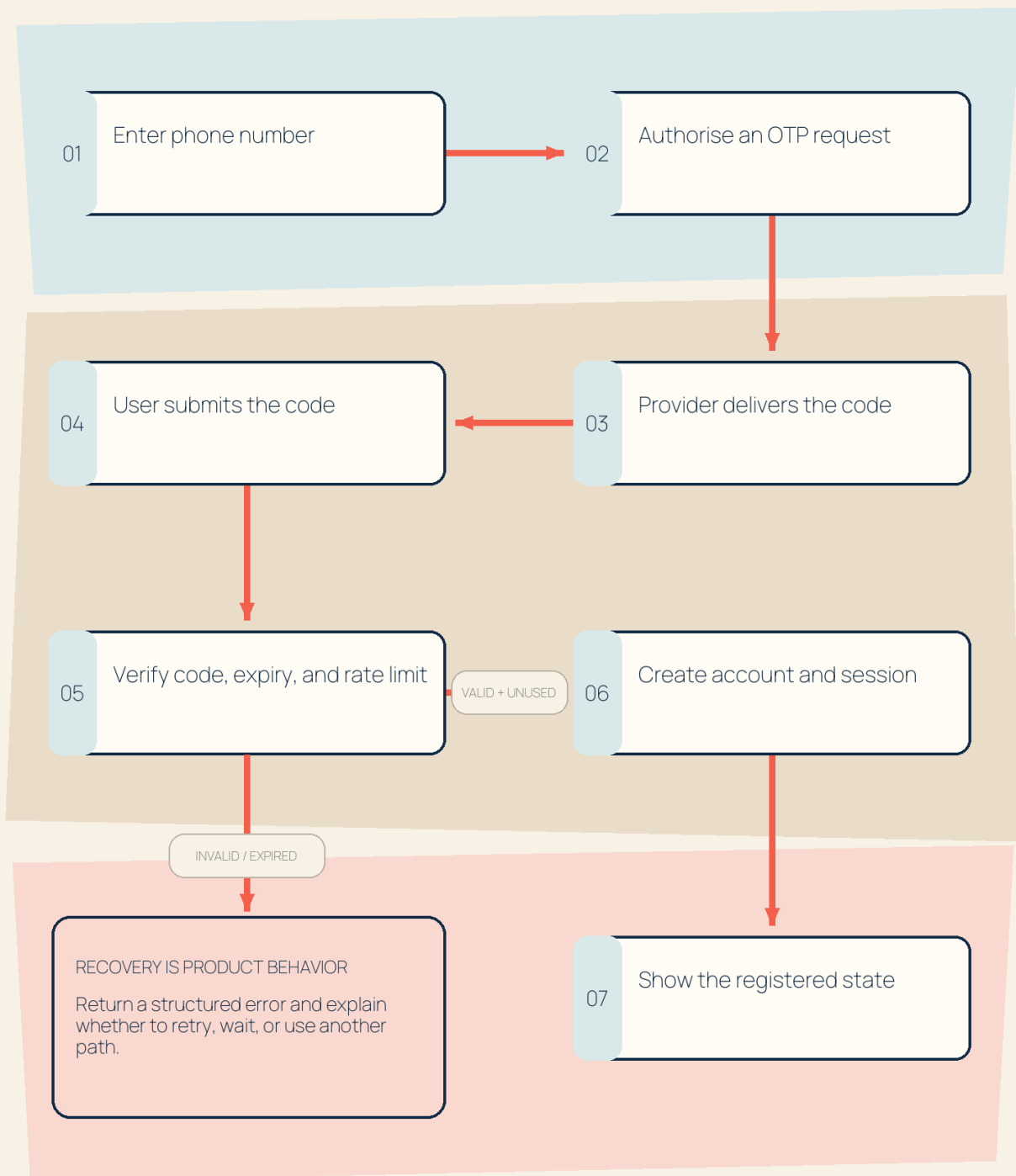
F1.1

Technical participation progresses through observable work, not status; the appropriate stopping point depends on the product and risk.

Figure 2.1. Registration request/response sequence

THE REGISTRATION PROMISE

One user action crosses product, identity, provider, storage, and recovery boundaries.



Success and recovery are two outcomes of the same registration promise.

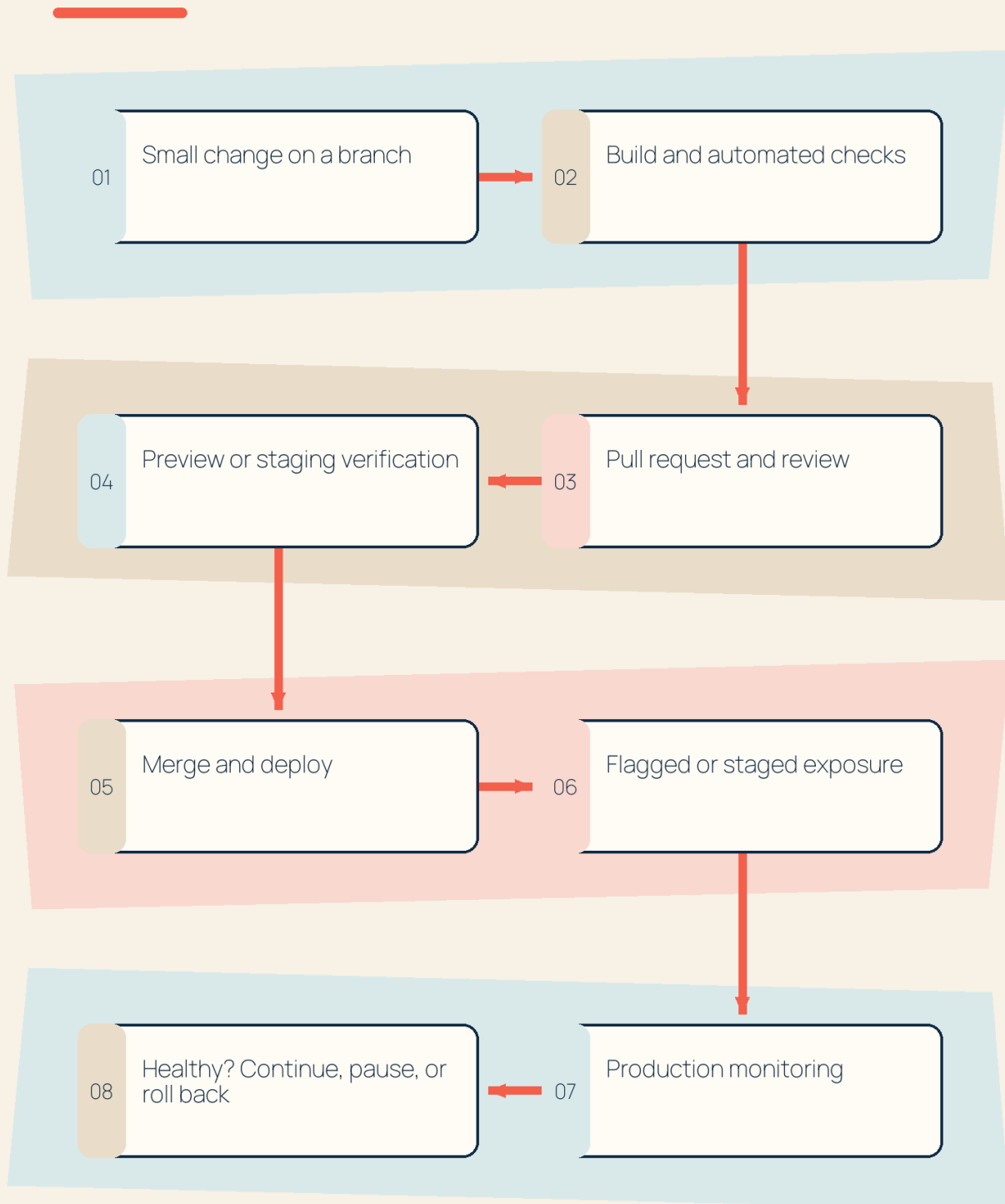
F2.1

The registration promise crosses multiple actors; success and recovery states both belong in the product conversation.

Figure 3.1. Commit-to-production delivery path

THE DELIVERY CONFIDENCE PATH

Integration, deployment, exposure, and learning are separate product decisions.



Shipping confidence accumulates one verified boundary at a time.

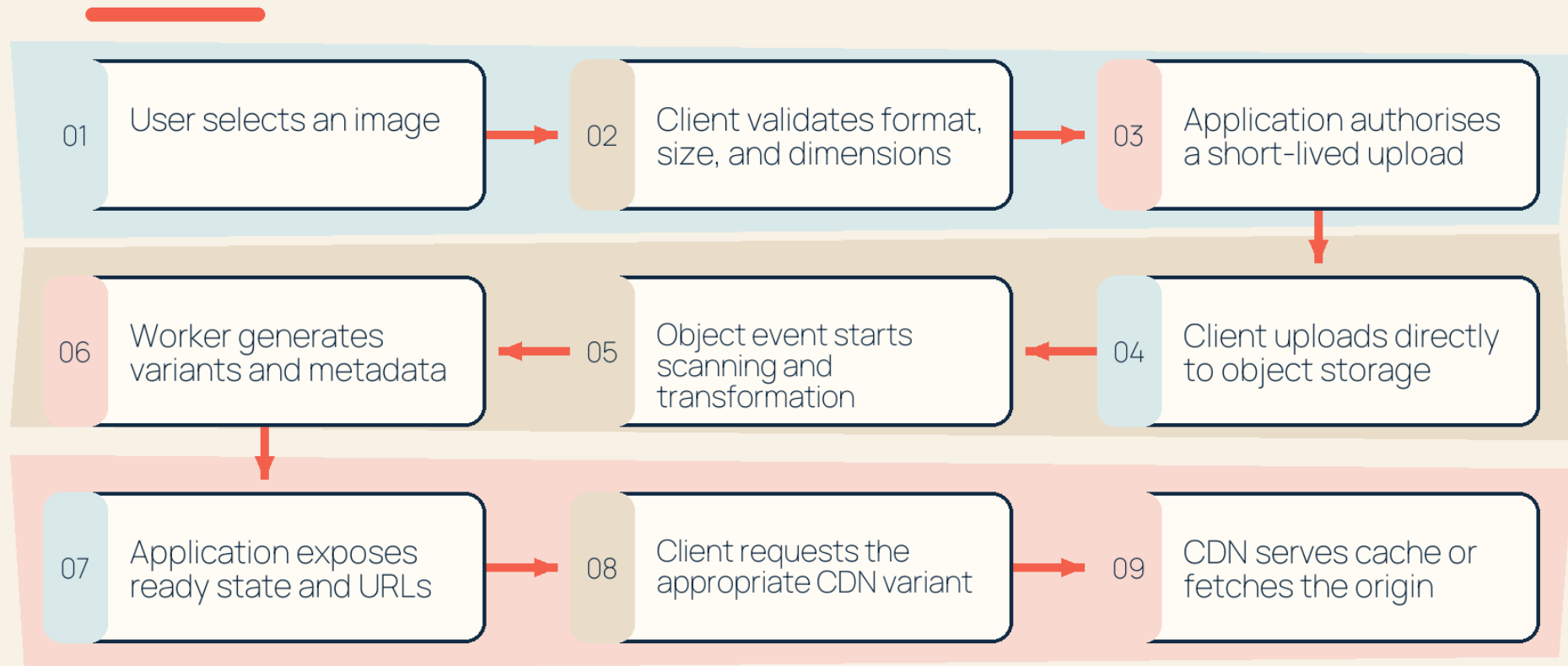
F3.1

A delivery path separates code integration, deployment, exposure, and the decision to continue.

Figure 4.1. Original asset to responsive delivery

THE IMAGE ASSET JOURNEY

The application authorises the action; specialised systems move and transform the file.



One user action crosses validation, authorisation, transfer, processing, metadata, and delivery boundaries.

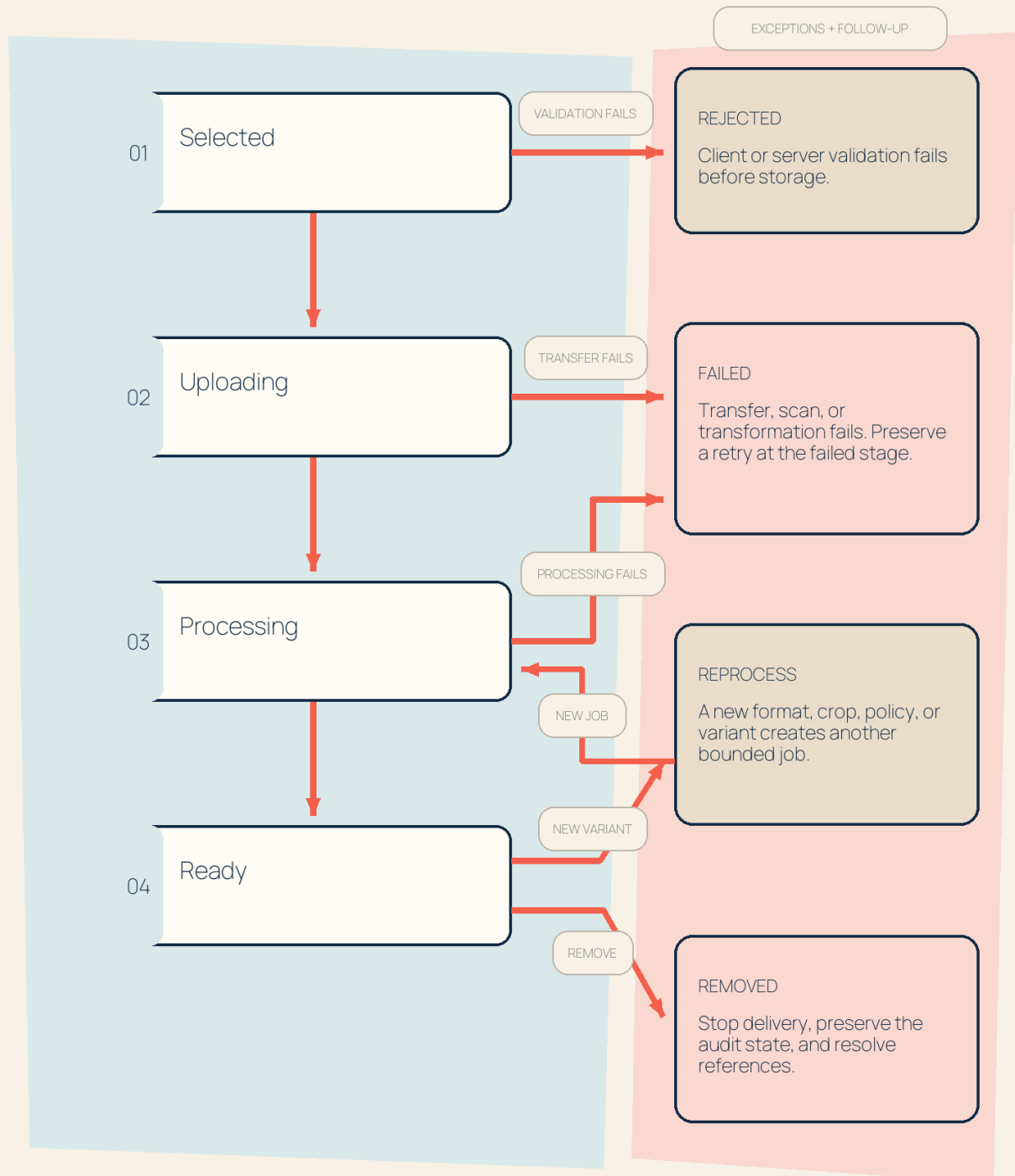
F4.1

The image-asset journey connects selection and validation to authorised transfer, processing, product state, variants, and delivery. Later sections examine each boundary.

Figure 4.2. Asset-state lifecycle

THE ASSET STATE MAP

Uploaded is not ready. Validation, transfer, processing, replacement, and removal are visible product states.



Every terminal-looking state still needs ownership and a user-visible next action.

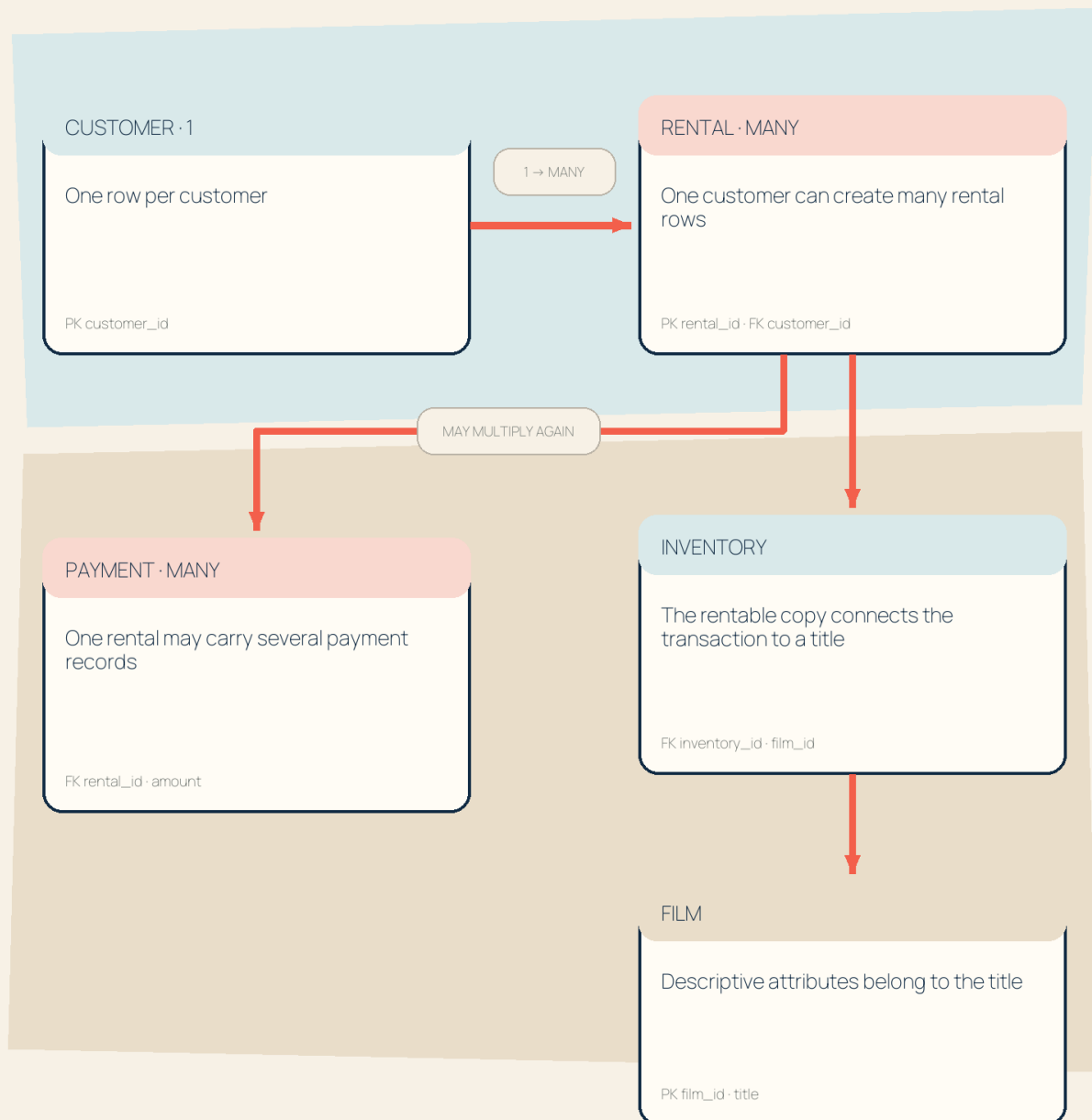
F4.2

Asset state is user-visible product behavior; "uploaded" is not the only meaningful outcome.

Figure 5.1. Relational grain through joins

HOW JOIN GRAIN EXPANDS

A correct join can still answer the wrong question when one business object becomes many rows.



NAME THE OUTPUT GRAIN BEFORE WRITING THE JOIN

Keys connect tables; the product question decides which row is the unit of analysis.

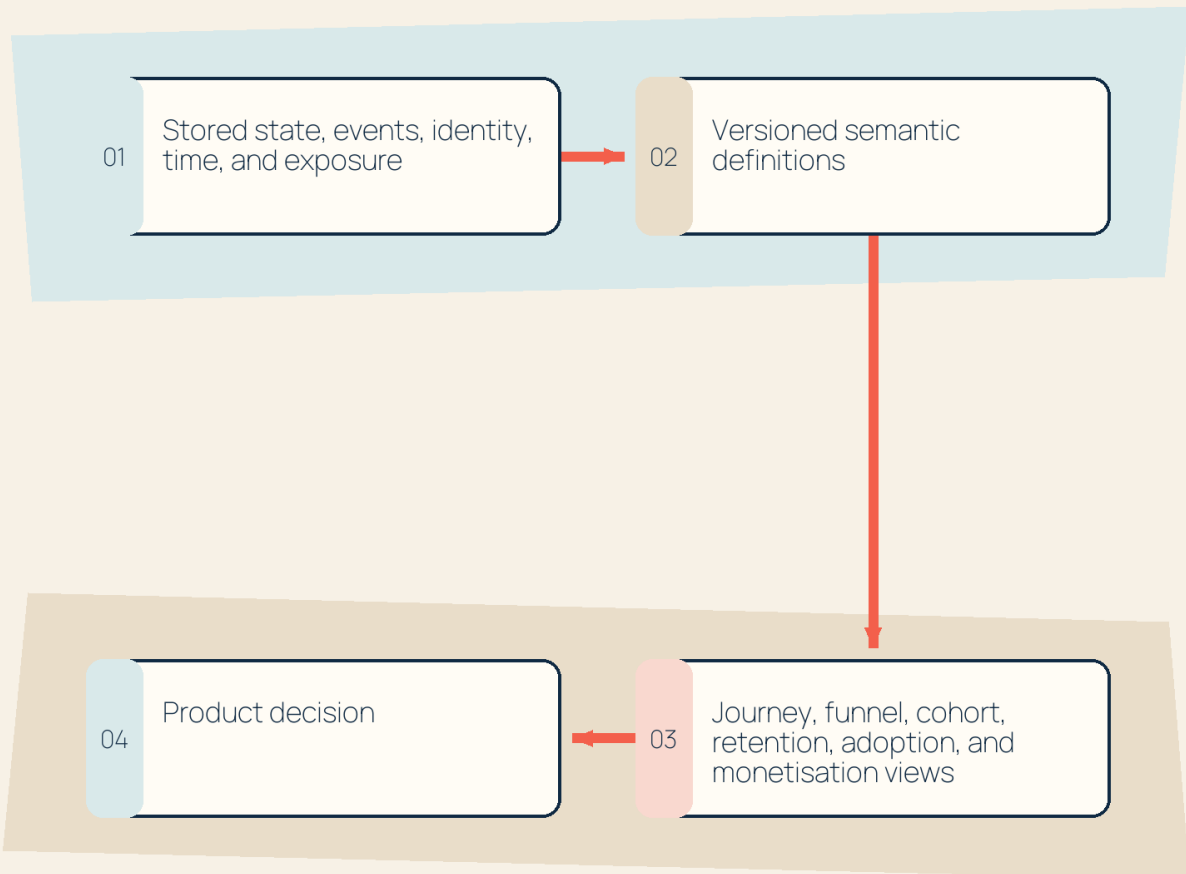
F5.1

The query grain changes as tables join: one customer can have many rentals, and one rental may have multiple payment records. Confirm the intended unit before aggregating.

Figure 6.1. Event-to-metric semantic chain

FROM PRODUCT REALITY TO A DECISION

Analytics is a governed interpretation layer, not an independent dashboard.



Every metric inherits the choices made beneath it.

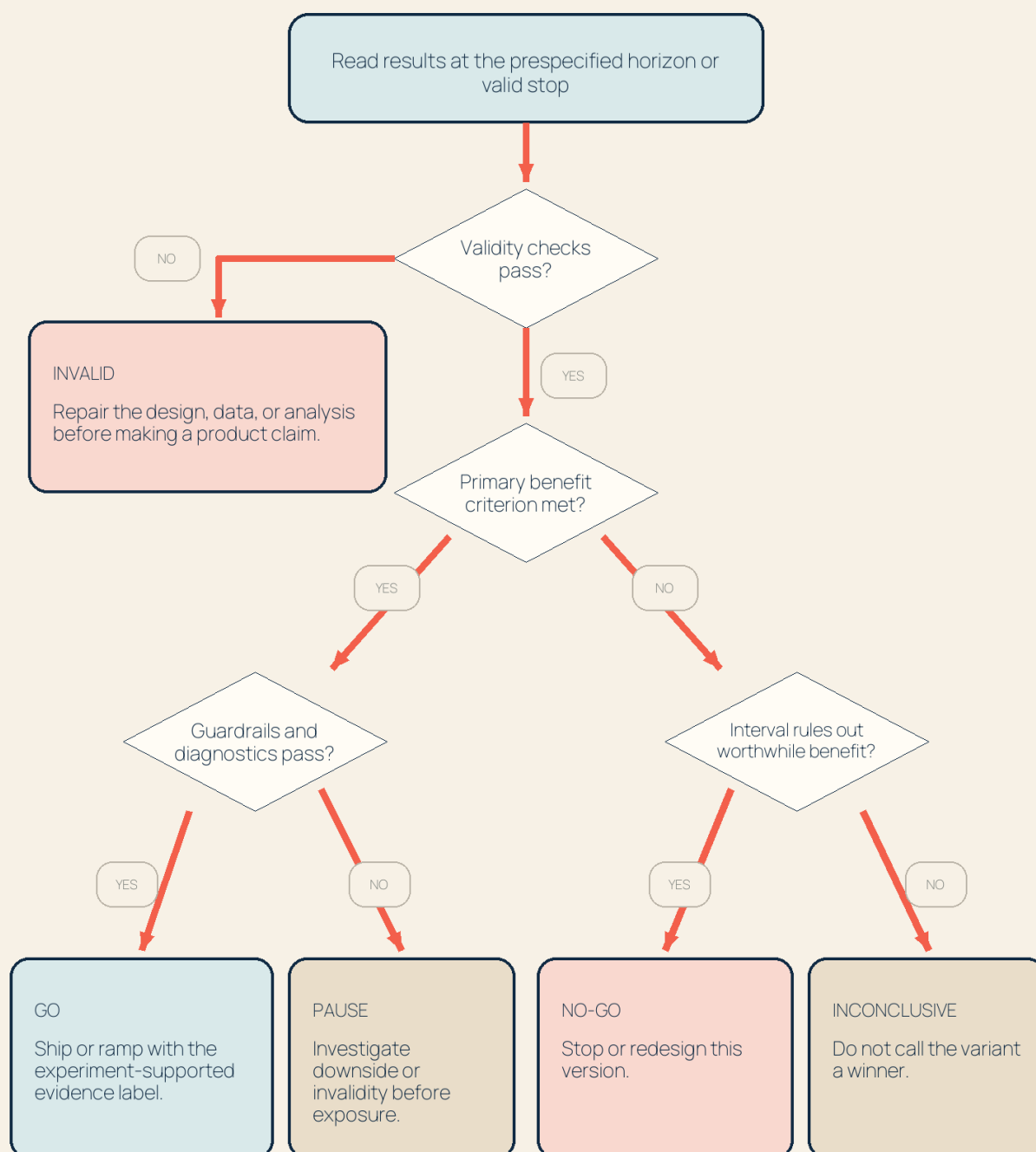
F6.1

Product analytics combines stored state, event streams, identity, time, and semantic definitions; no dashboard exists independently of those choices.

Figure 7.1. Experiment go/no-go decision flow

THE EXPERIMENT DECISION CANVAS

Evidence validity, product benefit, downside, and action are separate judgments.



A cautious rollout may still be a product decision, but the evidence label remains inconclusive.

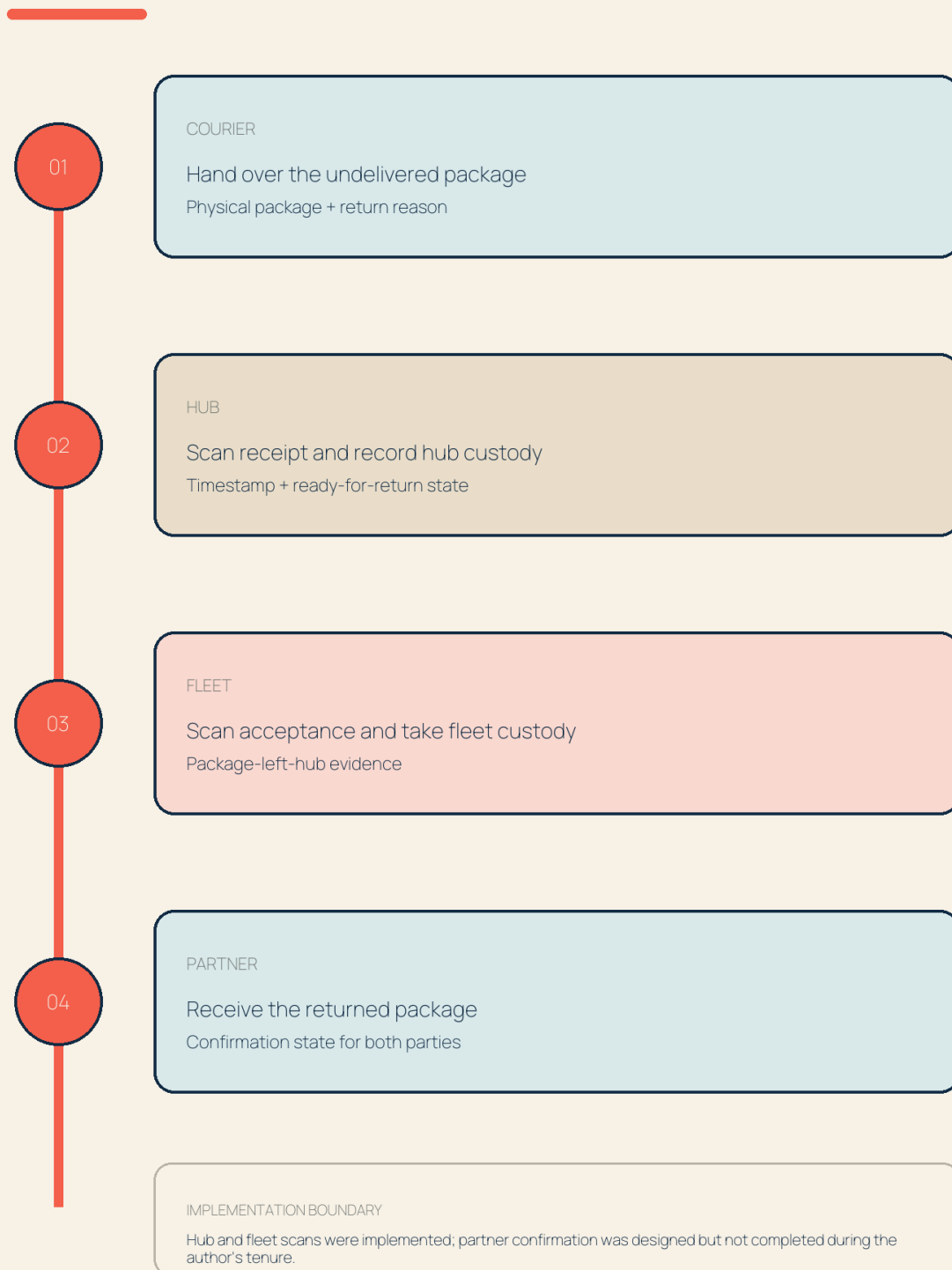
F7.1

The go/no-go flow separates evidence validity, product benefit, downside, uncertainty, and action.

Figure 8.1. Shipper reverse-logistics custody sequence

THE REVERSE-LOGISTICS CUSTODY TRAIL

A return becomes manageable when each physical hand-off leaves a timestamped system receipt.



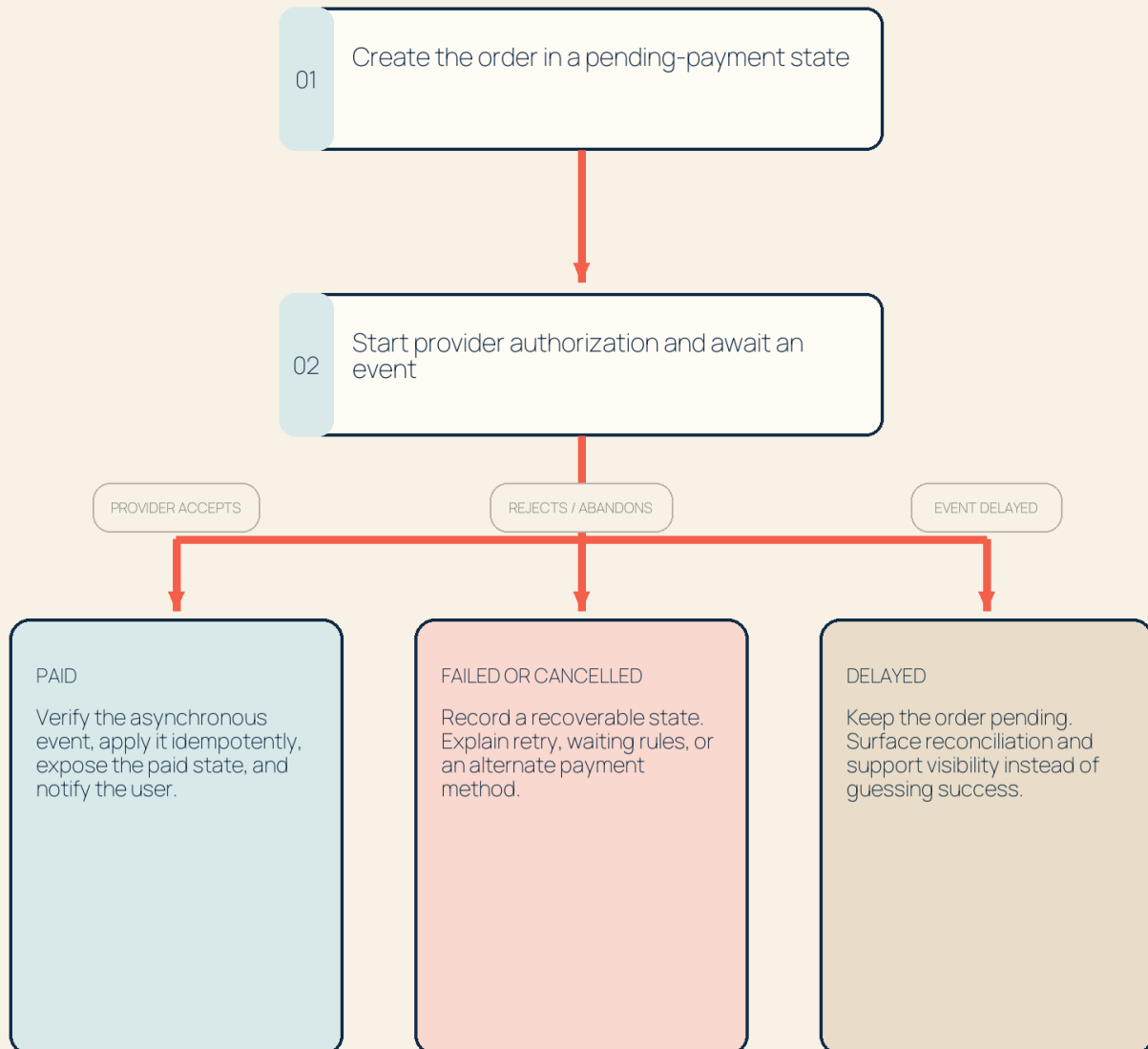
The backend does not move the package; it makes custody observable and disputable.

F8.1

Figure 8.2. E-wallet flow with happy, error, and recovery states

THE E-WALLET OUTCOME MAP

A payment requirement is incomplete until paid, failed, cancelled, and delayed states have product behavior.



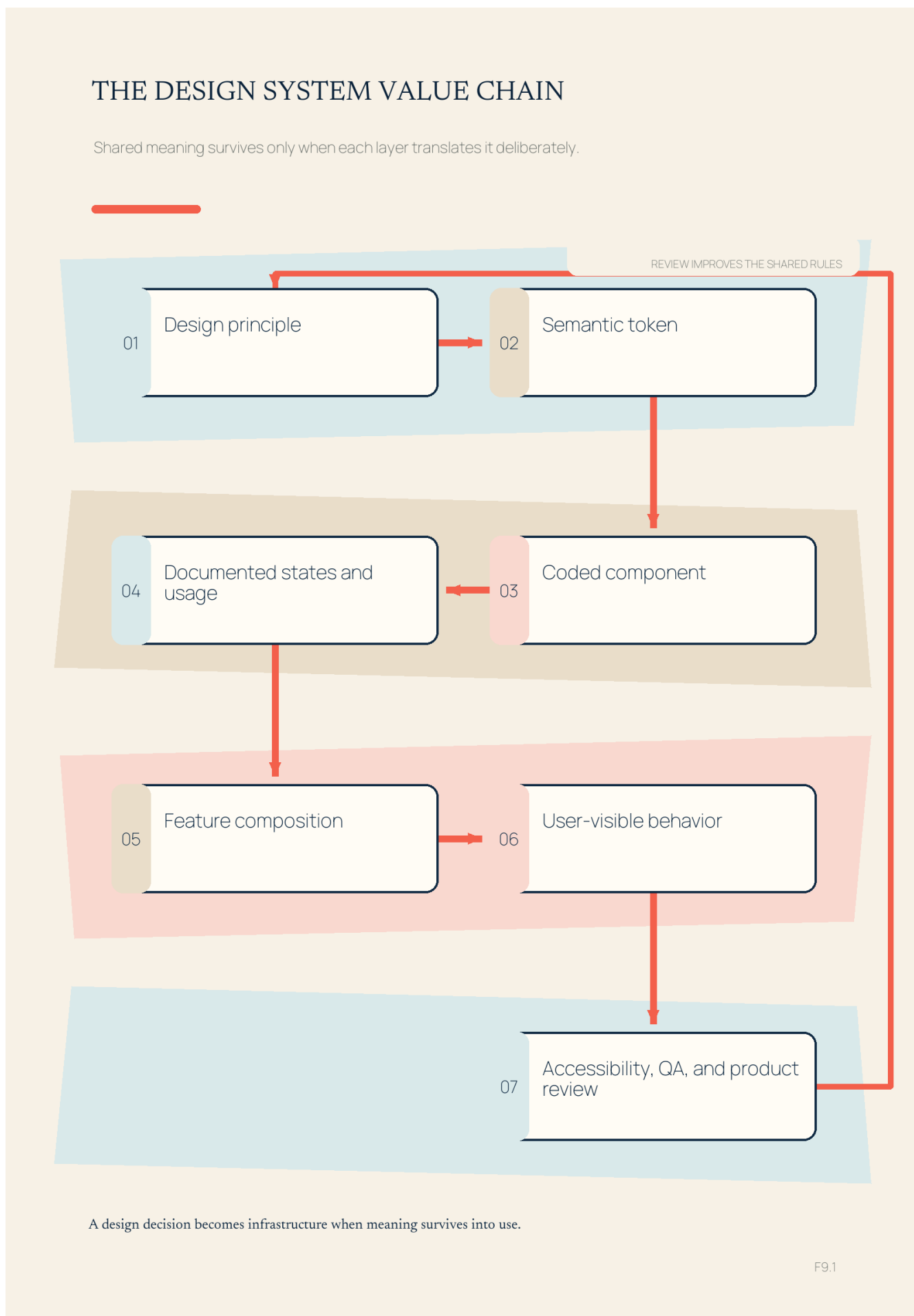
THE PROVIDER EVENT IS NOT THE USER EXPERIENCE

UI, order state, notification, operations, and support must agree on the same outcome.

F8.2

The e-wallet flow makes pending, failed, duplicate-safe, and support-visible behavior part of the requirement rather than late implementation detail.

Figure 9.1. Token-component-pattern-product hierarchy

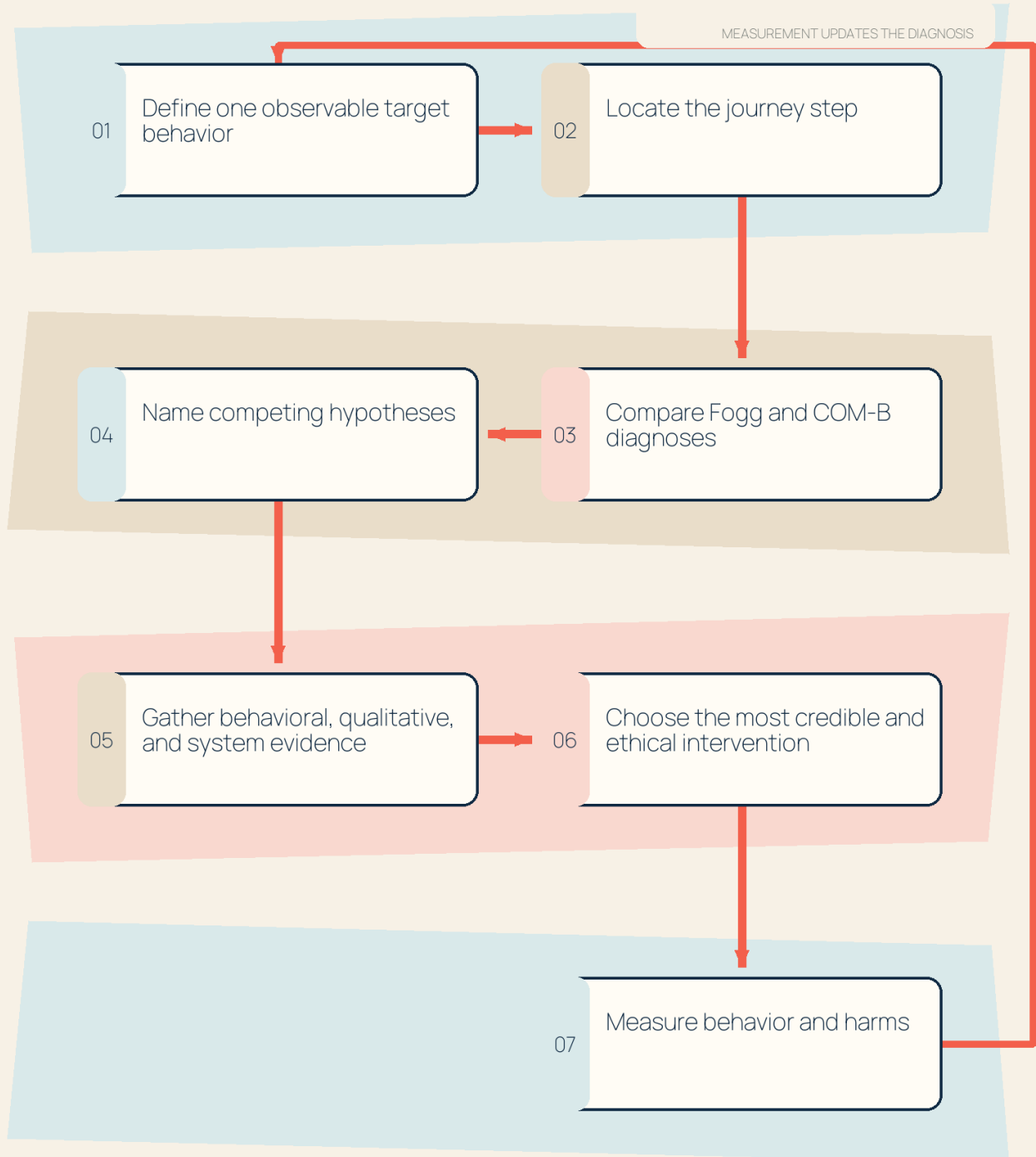


A design decision becomes infrastructure only when meaning survives from principle and token through code, states, composition, and review.

Figure 10.1. Behavior diagnosis: Fogg and COM-B

THE BEHAVIORAL DIAGNOSIS LOOP

Models produce competing explanations; evidence earns the intervention.



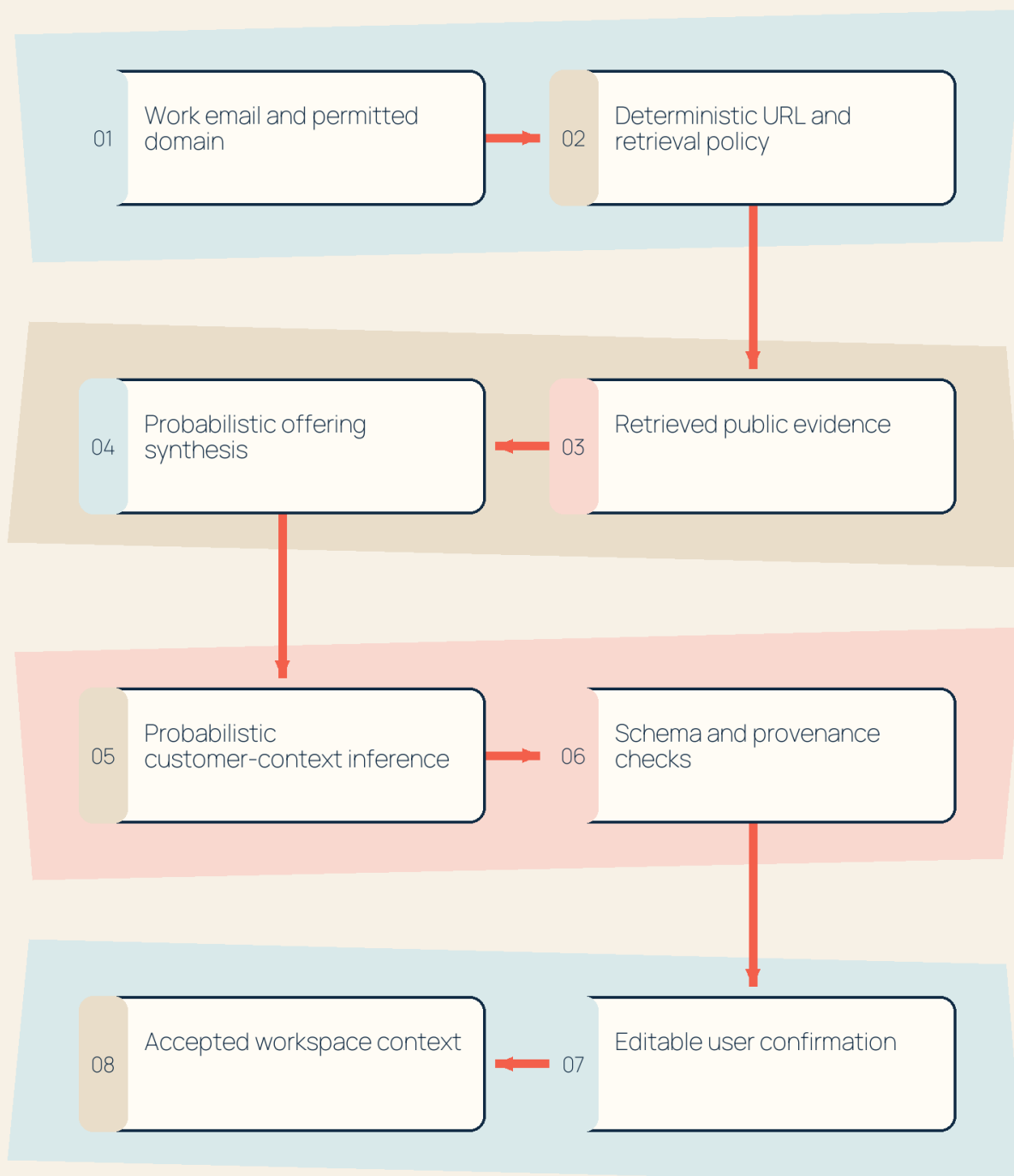
A plausible behavioral story is a hypothesis until evidence separates it from alternatives.

F10.1

Figure 11.1. Heatseeker onboarding deterministic–probabilistic boundary

THE GOVERNED CONTEXT PATH

Deterministic control surrounds probabilistic interpretation and preserves user authority.



When retrieval or validation fails, the product needs an explicit manual fallback.

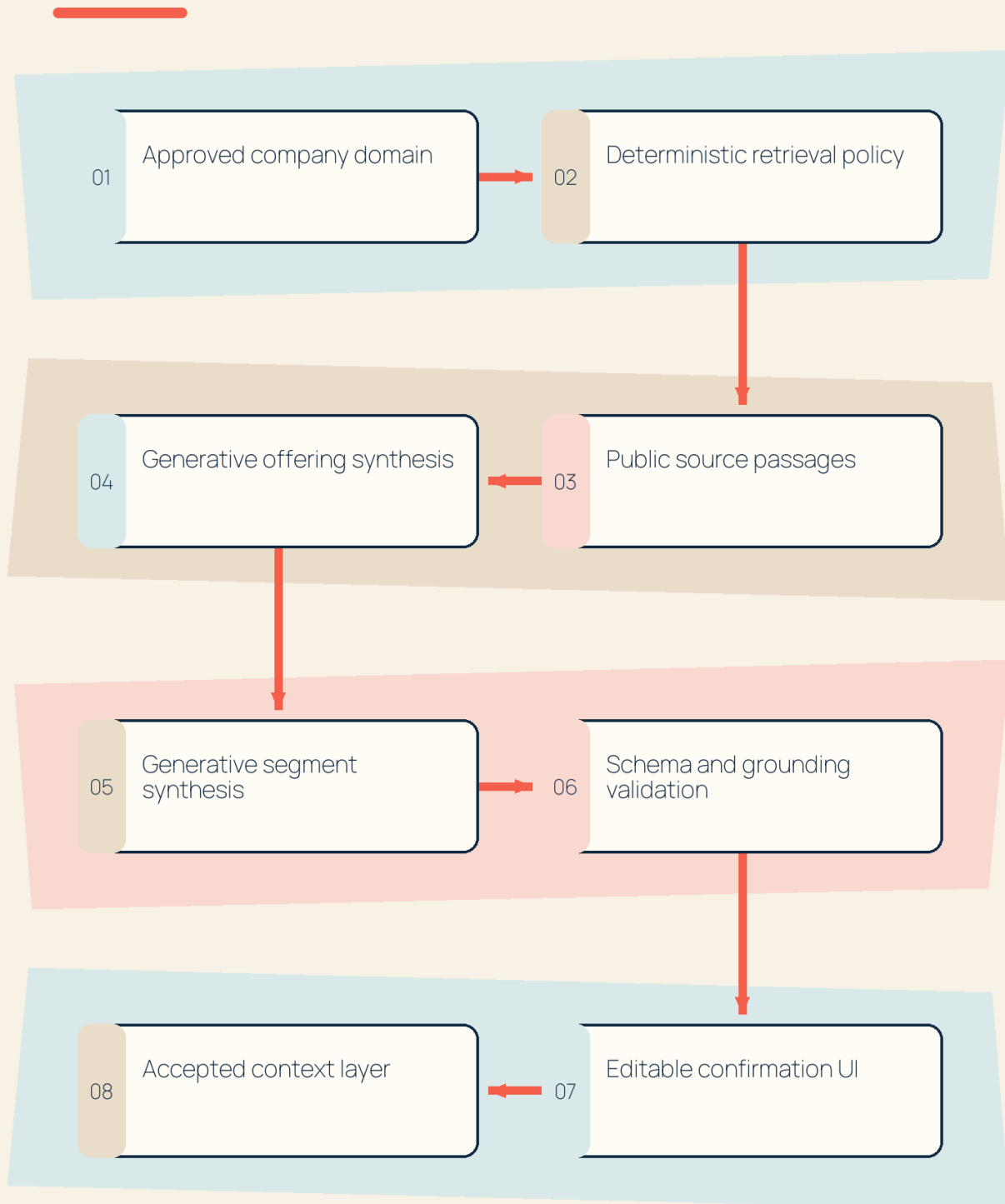
F11.1

The capability becomes governable when deterministic controls, probabilistic interpretation, validation, fallback, and user authority are explicit.

Figure 12.1. Heatseeker hybrid solution shape

THE HYBRID AI RESPONSIBILITY MAP

Exact control, bounded generation, validation, and user authority do different jobs.

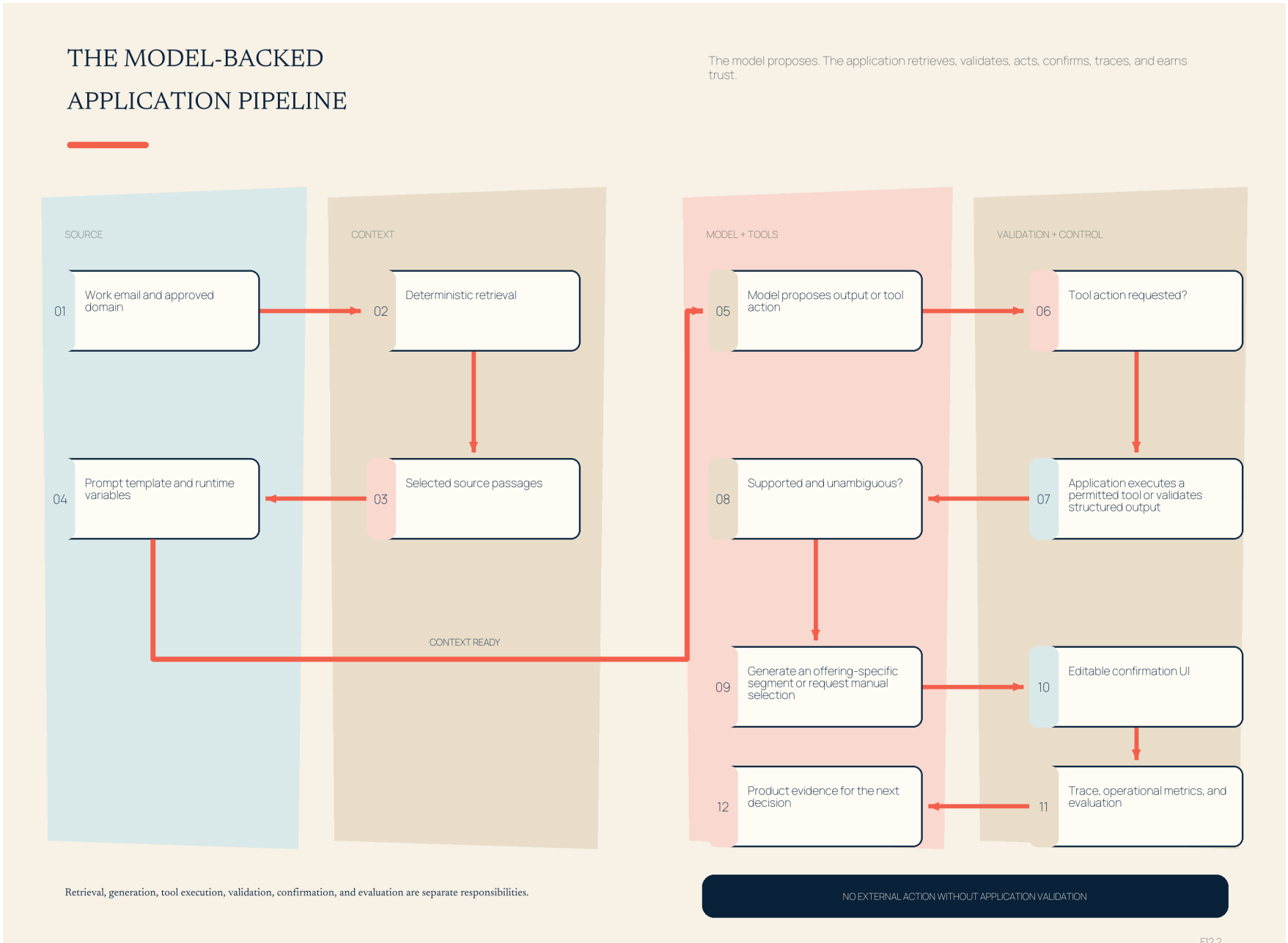


The uncertain component is bounded by controls before and after it.

F12.1

A hybrid design assigns exact control, bounded interpretation, validation, and user authority to different components.

Figure 12.2. Model-backed application pipeline

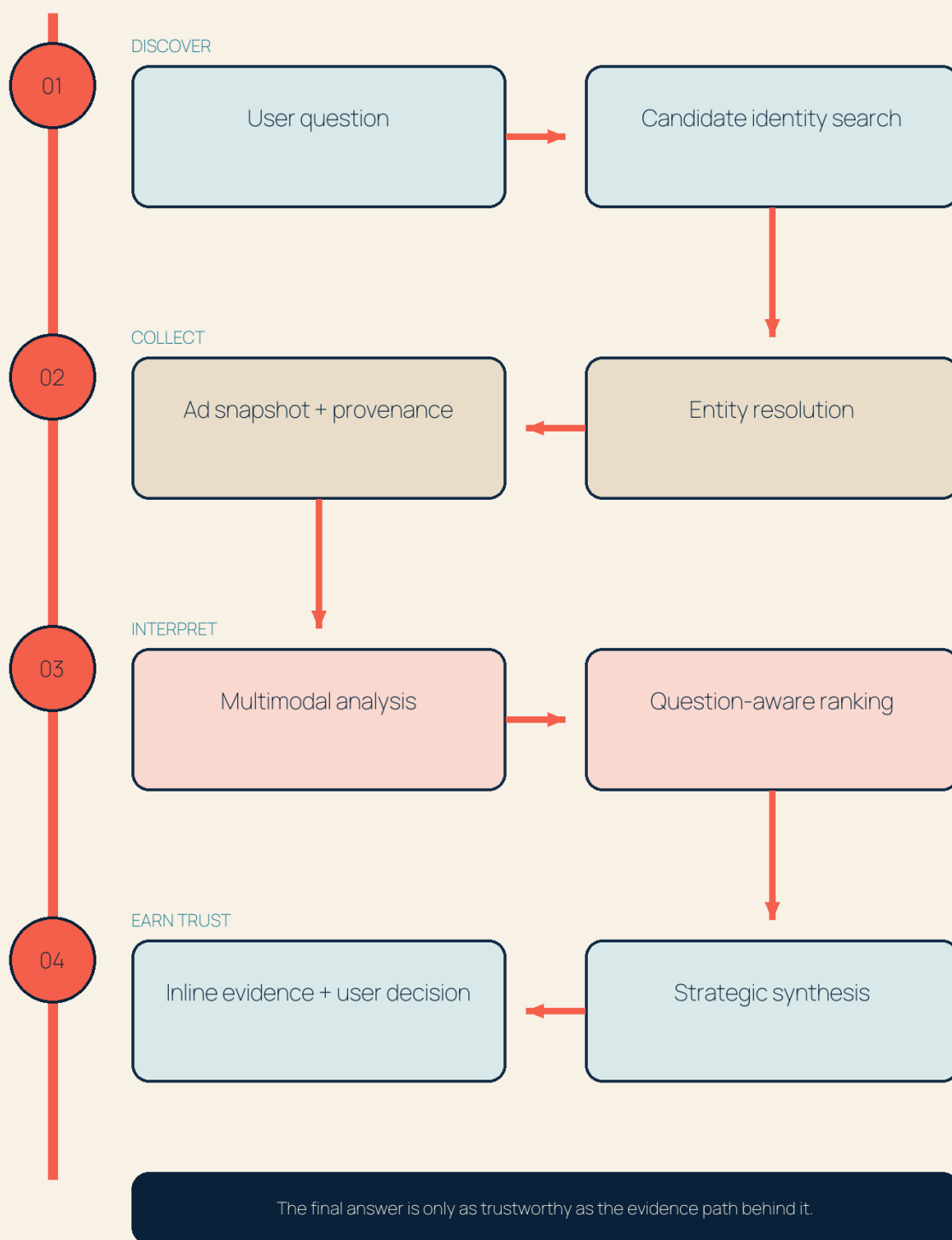


A model-backed feature is an application pipeline. Retrieval, context construction, generation, tool execution, validation, user control, tracing, and evaluation have different responsibilities.

Figure 13.1. Competitor-ad evaluation pipeline

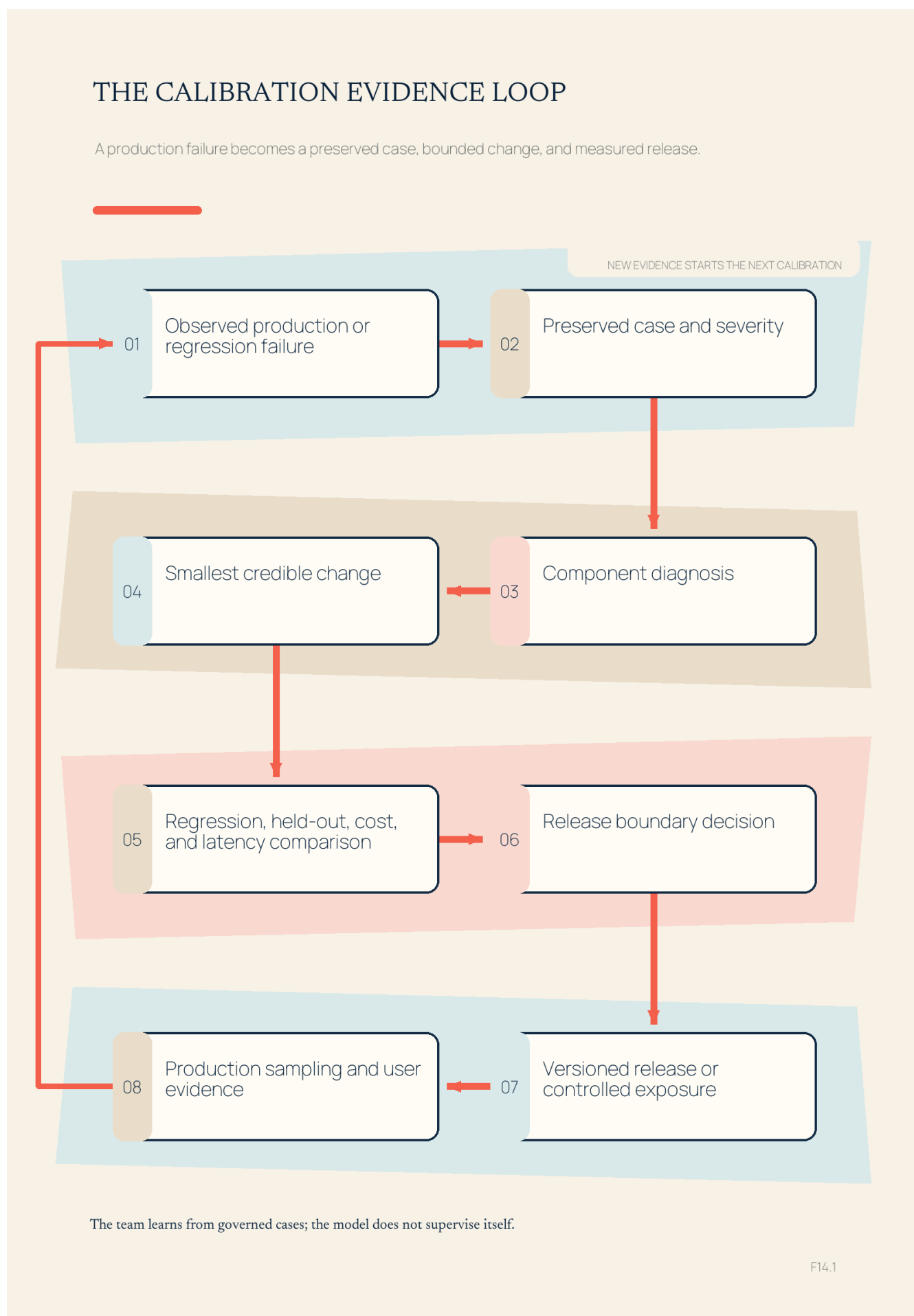
THE INSPECTABLE AI EVALUATION SYSTEM

Quality is produced across identity, evidence, analysis, ranking, synthesis, and user trust.



Final-answer quality depends on identity, evidence, analysis, ranking, synthesis, and UI trust—not only on the generation model.

Figure 14.1. Human-directed Heatseeker calibration loop

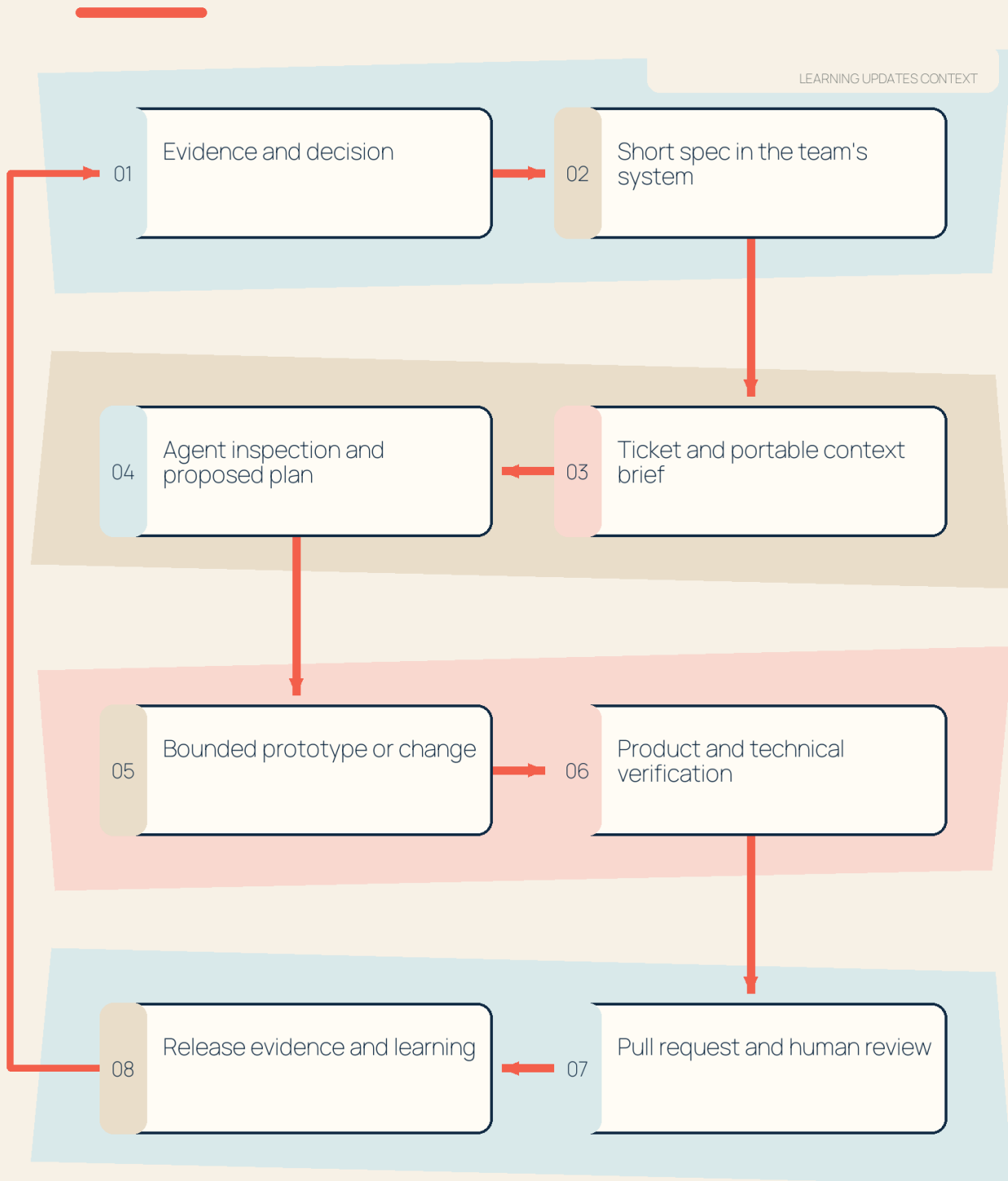


Calibration is a human-directed quality loop: an observed failure becomes a preserved case, bounded change, regression-tested release, and new production evidence.

Figure 15.1. Agent operating contract

FROM EVIDENCE TO VERIFIABLE WORK

Portable context keeps product intent intact across agents, artifacts, and review.



The useful unit of agent work is a verifiable contribution, not a prompt transcript.

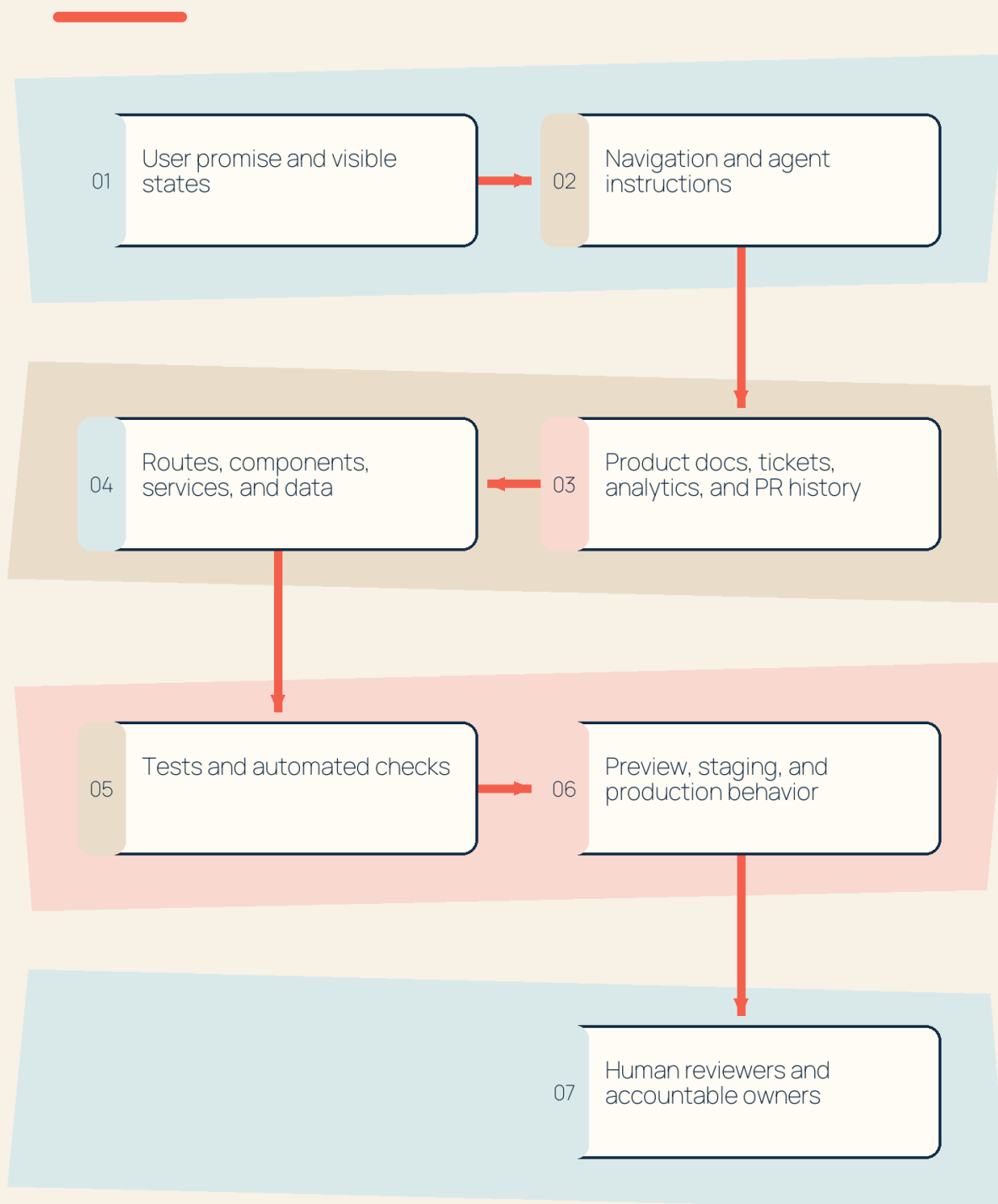
F15.1

Agent context should preserve the evidence-to-decision thread and end in human-verifiable work, not an isolated prompt transcript.

Figure 16.1. Repo as product surface

THE REPOSITORY ORIENTATION PATH

Start from the promise, then trace through instructions, implementation, checks, and ownership.

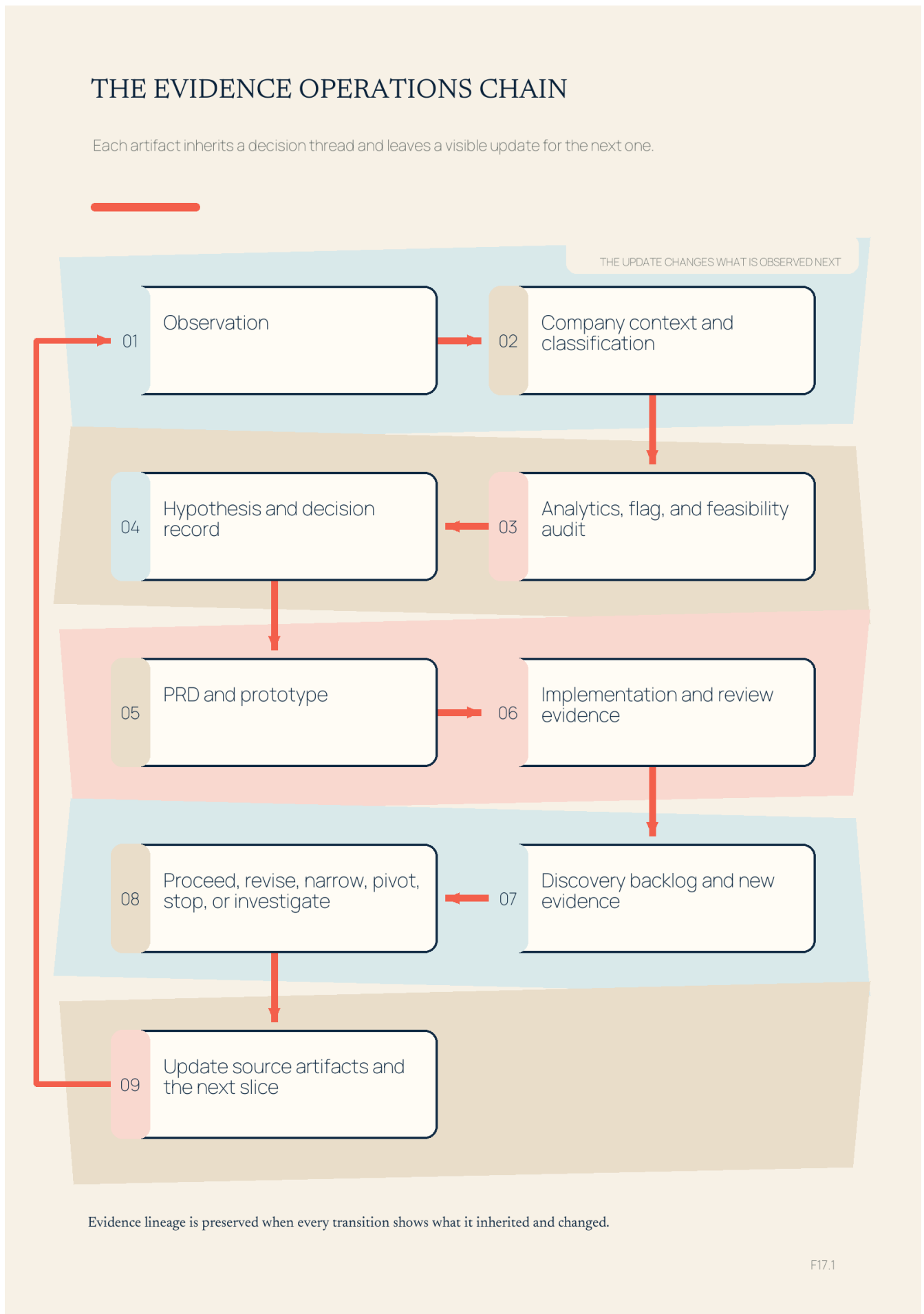


Product context explains intent; runtime evidence and accountable owners verify reality.

F16.1

Repo orientation connects the user promise to implementation, checks, environments, and accountable reviewers; product context can explain rationale, but it does not replace runtime verification or human ownership.

Figure 17.1. Evidence operations continuity

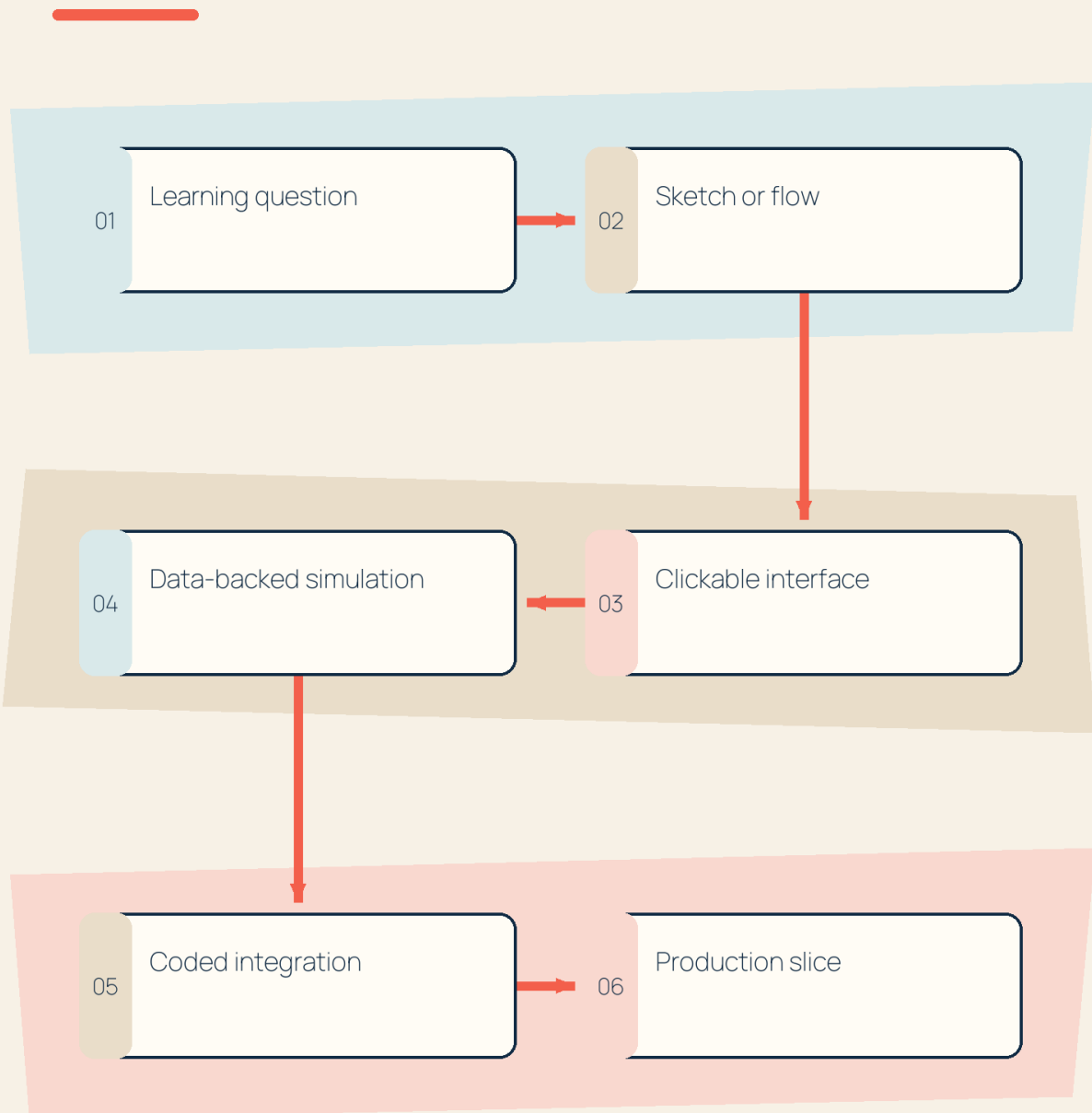


Evidence operations preserve lineage by carrying each observation through classification, product artifacts, implementation evidence, and an explicit decision that updates the next source of truth.

Figure 18.1. Prototype fidelity ladder

THE PROTOTYPE FIDELITY LADDER

Increase fidelity only when the learning question demands a more expensive proof.

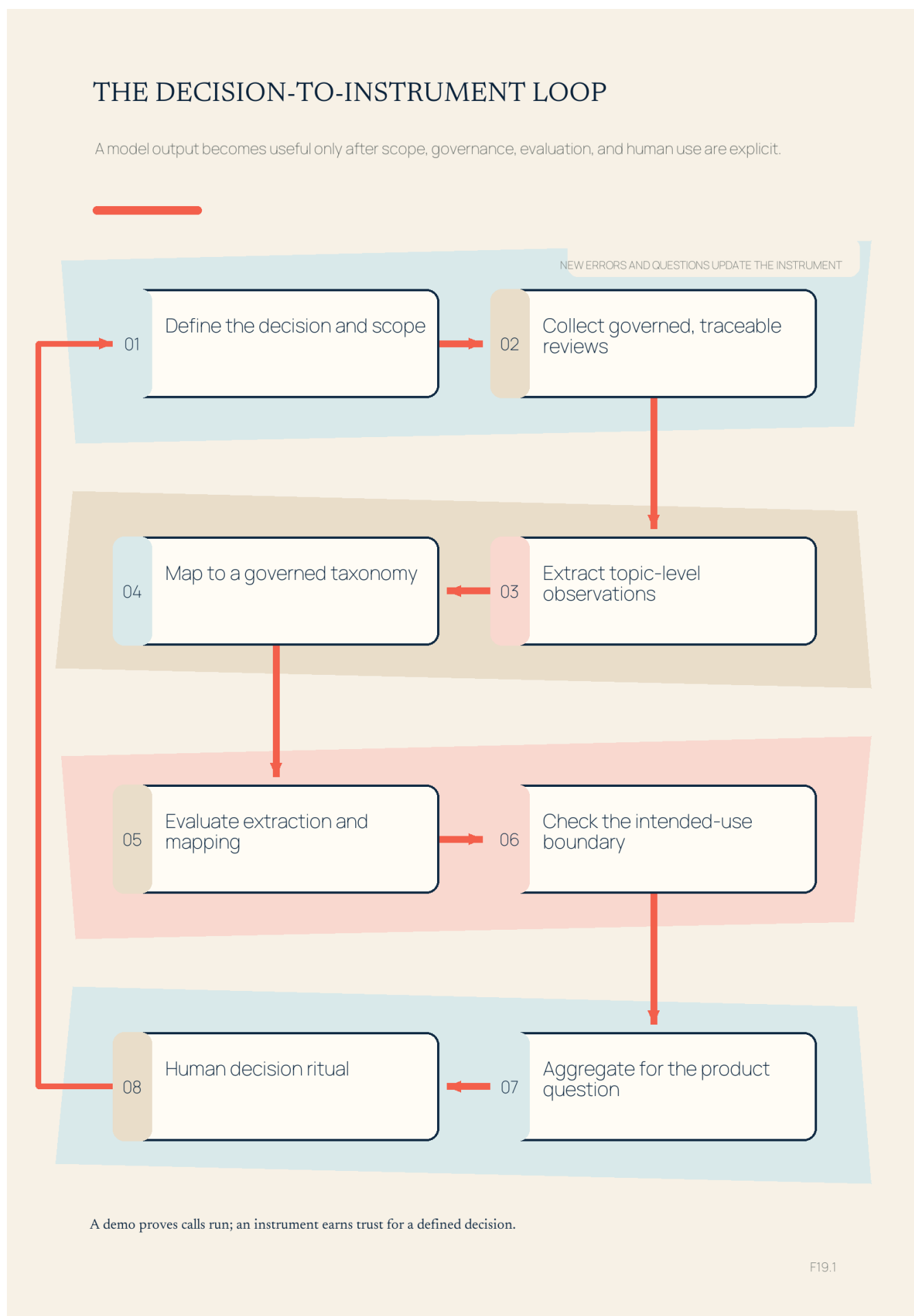


Prototype fidelity is a decision ladder, not a maturity race.

F18.1

Prototype fidelity is a decision ladder, not a maturity race: move only as far as the learning question requires, and treat production exposure as a separate commitment.

Figure 19.1. Decision-to-instrument loop

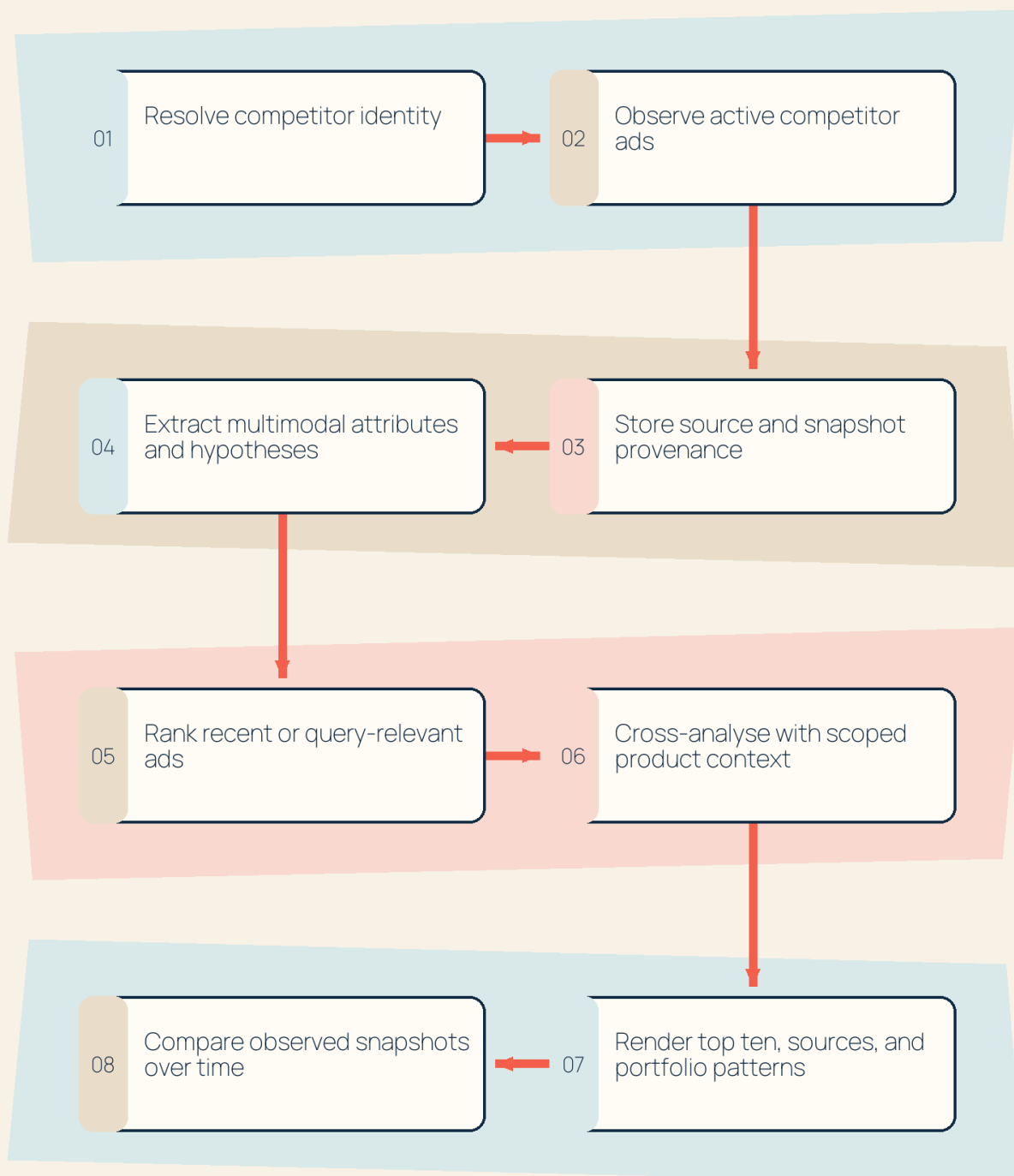


The original decision-to-instrument loop separates scope, governed collection, extraction, taxonomy, evaluation, aggregation, and human use so a model output does not become a dashboard without a quality boundary.

Figure 19.2. Competitor-ad transfer pipeline

THE COMPETITOR EVIDENCE PATH

Identity, provenance, multimodal interpretation, retrieval, and presentation stay separately inspectable.

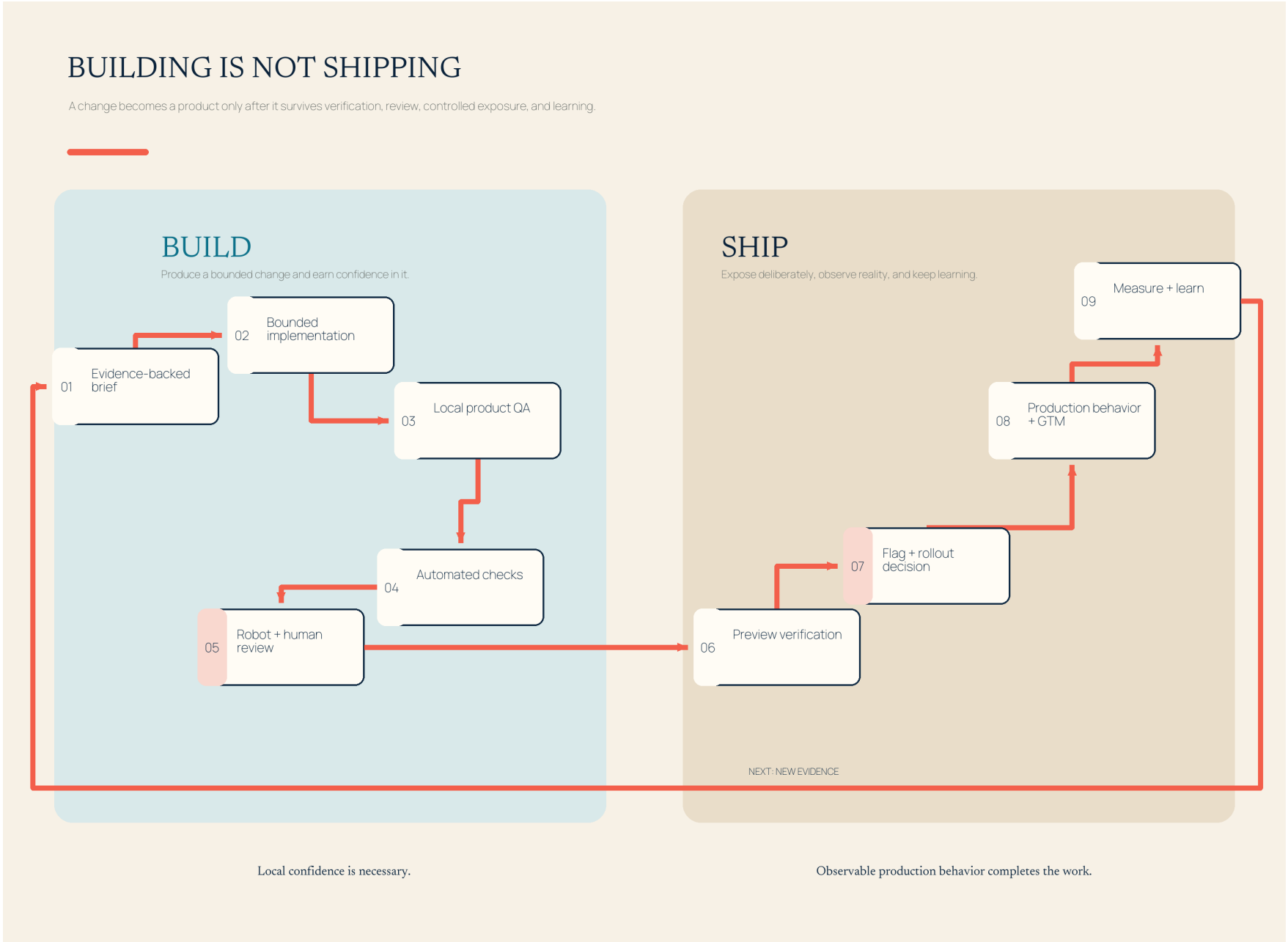


No creative enters the instrument until its advertiser identity is resolved.

F19.2

A competitor-ad response is trustworthy only when identity resolution, collection, provenance, extraction, retrieval, cross-analysis, and presentation remain separately inspectable.

Figure 20.1. Building is not shipping



Building becomes shipping only after the change survives verification, review, controlled exposure, and measurement.